2 08.30.23 - Metric Spaces

Definition 2.1. Let X be a set. Then $d : X \times X \to [0, \infty)$ is a **metric** if and only if for all $x, y, z \in X$,

1. $d(x, y) = 0$ if and only if $x = y$
2. $d(x, y) = d(y, x)$
3. $d(x, y) \leq d(x, z) + d(z, y)$ [triangle inequality]

Then (X, d) is a **metric space**.

Example 2.2. In $\mathbb{R}^n$, take

$$d(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i|^2 \right)^{\frac{1}{2}}$$

$$d(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{\frac{1}{p}}$$

Example 2.3. Let X be any set, take the *discrete metric*

$$d(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

Example 2.4. Let $S^1 = \{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$ be the unit circle. Then take

$$d(x, y) = \text{ arclength from } x \text{ to } y$$

$$d(x, y) = \text{ Euclidean distance as a subset of } \mathbb{R}^2$$

How do metric spaces transfer? A: Morphisms, specifically isomorphisms. The correct notion of isomorphism for metric spaces is an *isometry*.

Definition 2.5. Let (X, d_X), (Y, d_Y) be metric spaces, then $f : X \to Y$ is an **isometry** if it is bijective and satisfies $d_Y(f(x), f(x')) = d_X(x, x')$ for all $x, x' \in X$.

Idea: distance-preserving maps

Definition 2.6. Let (X, d_X) and (Y, d_Y) be metric spaces. $f : (X, d_X) \to (Y, d_Y)$ is **continuous at** $x \in X$ if and only if $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $\forall x \in X$,

$$d_X(x, x') < \delta \implies d_Y(f(x), f(x')) < \varepsilon$$

and is **continuous** if it is continuous at x for all $x \in X$.

In epsilon-ball notation, this is equivalent to $\forall \varepsilon > 0$, $\exists \delta > 0$ such that

$$f[B(x, \delta)] \subseteq B(f(x), \varepsilon)$$

Continuous maps make good morphisms in a relaxed sense. The isomorphisms arising from these are *homeomorphisms*.

Definition 2.7. $f : (X, d_X) \to (Y, d_Y)$ is a **homeomorphism** if f is continuous and f^{-1} exists and is continuous.

X, Y are then **homeomorphic**, written $X \cong Y$.

Question. What is preserved by continuous maps?

Example 2.8. Let $X = \{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$ be the unit circle and $Y = \{x \in \mathbb{R}^2 \mid |x_1| + |x_2| = 1\}$ be the diamond. Then X and Y are homeomorphic if we take

$$f : X \to Y \quad x \mapsto \frac{x}{|x|_1} = \frac{x}{|x_1| + |x_2|}$$

$$f^{-1} : Y \to X \quad y \mapsto \frac{y}{|y|_2} = \frac{y}{(y_1^2 + y_2^2)^{\frac{1}{2}}}$$

Hence corners are **not** topological.

Example 2.9. $f : \mathbb{R} \to (i\frac{\pi}{2}, \frac{\pi}{2}), x \mapsto \arctan(x)$ is a homeomorphism.

So boundedness is not topological. Neither is completeness.

Question. What is a topological property, if none of the above are?

Definition 2.10. Let (X, d) be a metric space. Then $U \subseteq X$ is **open** if and only if $\forall x \in U$, $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq U$.

Example 2.11. $\{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 < 1\}$ is open.

Definition 2.12. $C \subseteq X$ is **closed** if and only if $X \setminus C$ is open.

Theorem 2.13. $f : (X, d_X) \to (Y, d_Y)$ is continuous if and only if $f^{-1}[U]$ is open in X for every $U \subseteq Y$ open.

Proof. $\Rightarrow$ Suppose f is continuous and let $U \subseteq Y$ be open.

Claim. $f^{-1}[U]$ is open.

Let $x \in f^{-1}[U]$. Since $f(x) \in U$ and U is open, there is $\varepsilon > 0$ such that $B(f(x), \varepsilon) \subseteq U$. As f is continuous, there is $\delta > 0$ such that $f[B_\delta(x)] \subseteq B_\varepsilon(f(x)) \subseteq U$. So $B_\delta(x) \subseteq f^{-1}[U]$, so $f^{-1}[U]$ is open.

$\Leftarrow$ Let $x \in X$, $\varepsilon > 0$. Then $B_\varepsilon(f(X))$ is open, thus $f^{-1}[B_\varepsilon(f(x))$ is open and contains x. By the definition of open, $\exists \delta > 0$ such that $B_\delta(x) \subseteq f^{-1}[B_x f(x))] \implies f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. $\square$

3 09.01.23 - Topological Spaces

A *topological property* is a definition in terms of open sets. So continuity is a topological property.

Proposition 3.1. Let (X, d) be a metric space. Then

1. $\emptyset, X$ are open
2. finite intersections are open (if U, V are open, then $U \cap V$ is open)
3. arbitrary unions of open sets are open (if U_α for $\alpha in A$ (for arbitrary index set A), then $\bigcup_{\alpha \in A} U_\alpha$ is open)

Definition 3.2. Let X be a set, then τ, a set of subsets of X, is called a **topology** on X if

1. $\emptyset, X \in \tau$
2. if $U_1, \cdots, U_n \in \tau$, then $\bigcap_{i=1}^n U_i \in \tau$ (closed under finite intersections)
3. if $U_\alpha \in \tau$ for $\alpha \in A$, A arbitrary, then $\bigcup_{\alpha \in A} U_\alpha \in \tau$ (closed under arbitrary unions)

Set in τ are called **open** and (X, τ) is a **topological space** (or just **space**).

Example 3.3. Let $\tau = \{U \subseteq X \mid U \text{ open}\}$ where X is any metric space.

Example 3.4. Let $\tau = \{\varnothing, X\}$. This is called the *indiscrete topology*. This is **not** a metric space for $|X| \geq 2$.

Example 3.5. Conversely, the *discrete topology* is defined as $\tau = 2^X$ with the discrete metric.

Example 3.6. The *cofinite topology* is $\tau = \{U \subseteq X \mid X \setminus U \text{ finite or all of } X\}$. (Exercise: check that this is a topology!)

Example 3.7. Find all topologies on 3 points, say a, b, c, up to homeomorphism.

1. indiscrete: $\{\{a, b, c\}\}$
2. $\{\{a, b, c\}, \{a\}, \{b, c\}\}$
3. $\{\{a, b, c\}, \{a\}\}$
4. $\{\{a, b, c\}, \{a, b\}\}$
5. $\{\{a, b, c\}, \{a\}, \{a, b\}\}$
6. particular point topology (every set that contains the point is open, everything else is closed): x3 $\{\{a, b, c\}, \{a, b\}, \{b, c\}, \{b\}\}$

7. the powerset; note that the only metric space that comes from a finite set is the topology that comes from the discrete topology
8. $\{\{a, b, c\}, \{a, b\}, \{a\}, \{b\}\}$
9. $\{\{a, b, c\}, \{a, b\}, \{b, c\}, \{a\}, \{b\}, \{c\}\}$

Example 3.8. Let $f_n : [0, 1] \to [0, 1]$ continuous, $f : [0, 1] \to [0, 1], x \mapsto \lim_{n\to\infty} f_n(x)$ (*if it exists*; important!). This notion of convergence doesn't come from a metric, but it is induced by the *topology of pointwise convergence*.

Example 3.9. The *Zariski topology* is interesting in algebraic geometry; it is defined by its closed sets, which is kind of odd.

Definition 3.10. Let (X, τ_X) and (Y, τ_Y) be topological spaces. Then $f : X \to Y$ is **continuous** if $\forall U \in \tau_Y$, $f^{-1}(U) \in \tau_X$ and is **continuous at** $x \in X$ if $\forall U \in \tau_Y$ with $f(x) \in U$, $f^{-1}[U]$ contains a $V \in \tau_X$ with $x \in V$.

It is immediate that $f : X \to Y$ is continuous if and only if it is continuous at every $x \in X$.

We now extend the notion of homeomorphism from metric spaces to topological spaces.

Definition 3.11. $f : X \to Y$ is a **homeomorphism** if f is continuous and f^{-1} exists and is continuous. X and Y are then **homeomorphic**.

Example 3.12.

$$\{\{a, b, c\}, \{a\}\} \cong \{\{a, b, c\}, \{b\}\}$$

Clearly, f continuous and f^{-1} exists does **not** imply that f^{-1} continuous. f^{-1} existing implies that f is bijective, hence the existence of a bijection alone (which implies that the number of open sets is constant) is not sufficient, but with the preservation of inclusion, it is.

Example 3.13. Let $B_2 = \{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 < 1\}$ be the open disk. This is homeomorphic to $\mathbb{R}^2$.

The closed unit disk $\overline{B_2}$ is however **not** homeomorphic to $\mathbb{R}^2$; one has boundary points and is compact, the other is not.

The recurring theme here is that to tell spaces apart, we need to look for *obstructions*.

Definition 3.14. Let X be a topological space with fixed and an agreed-on τ, and let $U \subset X$. The **interior of** U is denoted

$$\operatorname{int}(U) = \bigcup_{\substack{V \in U \\ V \text{ open}}} V$$

the **closure of** U is denoted

$$\overline{U} = \bigcap_{\substack{C \geq U \\ C \text{closed}}} C$$

and the **boundary of** U is denoted

$$\partial U = \overline{U} \setminus \operatorname{int}(U)$$

Remark. You can define a topology with the closure operator. (Exercise: check!)

4 09.06.23 - New Spaces from Old

In particular, *subspaces* and *product spaces*.

Definition 4.1. $A \subseteq X$ with $\tau_A = \{U \cap A : U \in \tau\}$ is a **subspace** and τ_A is called the **subspace topology**.

This makes $\iota : A \to A, x \mapsto x$ (the inclusion) continuous, and this is the inclusion-minimum topology with this property, i.e. if τ' is another topology on A such that $(A, \tau') \hookrightarrow (X, \tau)$ is continuous, then $\tau_A \subseteq \tau$.

Example 4.2. S^2

Definition 4.3. Let X, Y be topological spaces. If $f : X \to Y$ is injective and a homeomorphism onto its image, it is called an **embedding**.

X is then a subset of Y by f, so $f[X]$ carries the subspace topology with respect to Y.

Question. Given two spaces, can you define one as a subset of the other?

If yes, we can simply define a homeomorphism. If not, we are in more trouble...

Definition 4.4. LeT X be a set, Y_j be topological spaces, and $f_j : X \to Y_j$ be functions for $j \in J$ for some arbitrary J. The **initial topology** on X is the **coarsest** topology where all f_j are continuous.

It is easily verified that if τ_i for $i \in I$ are topologies in X, then $\bigcap_{i \in I} \tau_i$ is **also** a topology on X.

Note that the subspace and product topologies are special cases of the initial topology, and the final topology (which is the finest such topology) exists.

Definition 4.5. If τ_1 and τ_2 are topologies on X with $\tau_1 \subseteq \tau_2$, then τ_1 is **coarser** than τ_2 and τ_2 is **finer** than τ_1.

Proposition 4.6. There hence exists some unique **coarsest** topology by tkaing the intersections of all topologies.

Definition 4.7. Let X be a set, Y_j be topological spaces, and $f_j : X \to Y_j$ for $j \in J$ be functions. Then τ is the initial topology with regard to $\{f_j \mid j \in J\}$ if and only if

for any topological space Z and

$$\begin{tikzcd} Z \arrow[d, "g"] \arrow[dr, "f_j \circ g"] & \\ X \arrow[r, "f_j"] & Y_j \end{tikzcd}$$

then g is continuous if and only if all $f_j \circ g$ are continuous.

Note that one topology can be initial for two classes of functions - for example, just add more functions.

Theorem 4.8. The initial topology exists and is unique.

Proof. **Existence:** Note that the discrete topology on X makes all the f_j continuous, so we now have at least one topology. To find the coarsest, among all topologies τ_i for $i \in I$ that make all f_j continuous, one is coarser than the others, $\bigcap_{i \in I} \tau_i \subseteq \tau_i, i \in I$ and makes all f_j continuous, since for $U \in X$ open, $f_j^{-1}(U) \subseteq X$ is a set in τ_i for any $i \in I$, which implies that $f_j^{-1}[U] \in \bigcap_{i \in I} \tau_i$.

Uniqueness: If τ_1, τ_2 are initial with regard to $\{f_j\}$, then $\tau_1 = \tau_2$. Notice that $\tau_1 \cap \tau_2$ is also initial with regard to $\{f_j \mid j \in J\}$. Since f_j is continuous, so $(X, \tau_2) \to (X, \tau_1), x \mapsto x$ is continuous, which implies $\tau_1 \subseteq \tau_2$, and vice versa. Hence we achieve equality.

Alternatively, we can define the initial topology as the topology generated by the sets $f_j^{-1}[U], U \subseteq Y_j$ open (by taking finite intersections and arbitrary unions). $\square$

Definition 4.9. Let X be a topological space and $\{A_i \mid i \in \tau\}$ be subsets of X. Then $\{A_i \mid i \in I\}$ is called the **subbasis** or **generating set** if the topology on X is generated by $\{A_i \mid i \in I\}$.

Definition 4.10. $\{A_i \mid i \in I\}$ is a **basis** for the topology of X is the topology of X consists of all arbitrary unions of A_i.

Remark. The subbasis generates the basis.

Example 4.11. $\mathbb{R}^2$ has basis $\{B_\varepsilon(x) \mid \varepsilon > 0, x \in \mathbb{R}^2\}$

$\mathbb{R}^2$ has subbasis $\{\mathbb{R} \times (a, b), (a, b) \times \mathbb{R} \mid a, b, \in \mathbb{R}\}$

Definition 4.12. Let X and Y be topological spaces. The initial topology on $X \times Y$ with regard to $\pi_X : X \times Y \to X, (x, y) \mapsto x$ and $\pi_Y : X \times Y \to Y, (x, y) \mapsto y$ is called the **product topology**.

Thus $U \times Y$ for U open in X and $X \times V$ for V open in Y is a subbasis for the product topology, and $U \times V$ for U open in X and V open in Y is a basis for the product topology.

5 09.08.23 - Connectedness

Example 5.1. $\{0,1\}^{\mathbb{N}}$. The product topology on $\{0,1\}^{\mathbb{N}}$ is **not** discrete even though all $\{0,1\}$ have the discrete topology.

Let $\pi_j : \{0,1\}^{\mathbb{N}} \to \{0,1\}, (x_i)_i \mapsto x_j$. Then $\pi_j^{-1}(\{0\}) = \{0,1\} \times \{0,1\} \times \cdots \times \{0\}$(this is the jth coordinate) $\times \cdots \times \{0,1\} \times \cdots$. This is similar with $\pi_j^{-1}(\{1\})$.

All nonempty open sets in $\{0,1\}^{\mathbb{N}}$ have the form $A_1 \times A_2 \times A_3 \times \cdots$, where $A_i = \{0\}$ finitely many times, $A_i = \{1\}$ finitely many times, and otherwise, $A_i = \{0,1\}$. Hence open sets are *HUGE* by definition.

Note that the complements of these sets are also open, so these sets are clopen (closed and open).

Example 5.2. This is the familiar product from doing $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$.

Example 5.3. Recall that $S^1 = \{x \in \mathbb{R}^2 \mid \|x\| = 1\}$ is the circle. Then $S^1 \times S^1$ is the torus (this is naturally a subspace of $\mathbb{R}^4$, but you can also put this in $\mathbb{R}^3$).

We now try to classify spaces that embed into some product of $[0,1]$.

Definition 5.4. Let X be a topological space. X is **connected** if whenever $X = U \cup V$ for U, V open and $U \cap V = \emptyset$, one of U or V is empty.

A space is **disconnected** if $X = U \cup V$ for some U, V open such that $U \cap V = \emptyset$ with U and V nonempty (U, V open, disjoint, nonempty).

Example 5.5. Intervals in $\mathbb{R}$ are all connected sets in $\mathbb{R}$.

Proof. If I is not an interval, there are $x, y \in I$ and z with $x < z < y$ such that $z \notin I$. Then note that $I = ((-\infty, z) \cap I) \cup (I \cap (z, \infty))$, both of which are open in I as a subspace of $\mathbb{R}$ and nonempty because $x \in I \cap (-\infty, z)$ and $y \in I \cap (z, \infty)$, and they are disjoint as $(-\infty, z) \cap (z, \infty) = \emptyset$. So by contraposition, if I is connected, then I is an interval.

Now we show that intervals are connected. Suppose that $[a, b] = U \cup V$ for U, V nonempty disjoint open sets. Let $s = \sup(U)$ and WLOG suppose $b \in V$. Now, $s \notin U$ since otherwise there exists some $\varepsilon > 0$ such that $(s - \varepsilon, s + \varepsilon) \subseteq U$. But similarly, $s \notin V$, which is a contradiction. Hence $[a, b]$ is connected. $\square$

We can show that connectedness is a topological property.

Proposition 5.6. Let X, Y be topological spaces, $f : X \to Y$ be continuous, and X be connected. Then $f[X]$ is connected.

i.e. the image of connected sets is connected

Proof. Let $f(x) = U \cup V$ for U, V open and $U \cap V = \emptyset$. Then $X = f^{-1}[U] \cup f^{-1}[V]$, which are both open by the definition of continuity, and $f^{-1}[U] \cap f^{-1}[V] = \emptyset$.

Since X is connected, this implies that $f^{-1}[U] = \emptyset$ or $f^{-1}[V] = \emptyset$, thus U or V is empty. $\square$

Remark. Preimages commute with intersections and unions.

Proposition 5.7. Let X be a topological space. Then X is connected if and only if for every continuous map $f : X \to \mathbb{R}$, the image $f[X]$ is connected.

Proof. $\Rightarrow$ is the previous proposition.

$\Leftarrow$ Let $X = U \cup V$ for U, V open and disjoint. Then take $f : X \to \mathbb{R}, x \mapsto \begin{cases} 1 & x \in U \\ 0 & x \in V \end{cases}$. Since $f[X]$ is an interval, then f must be constant, which implies that $U = \emptyset$ or $V = \emptyset$ and hence X is connected. $\square$

Definition 5.8. Let X be any space. Define an equivalence relation on X by $x \sim y$ if there is a connected $C \subseteq X$ with $x, y \in C$.

We can check this equivalence relation:

1. $x \sim x$, shown by $C = \{x\}$
2. if $x \sim y$ then $y \sim x$ since containment doesn't depend on order
3. if $x \sim y$ and $y \sim z$ then $x \sim z$ since if there exists a connected C with $x, y \in C$ and C' with $y, z \in C'$, $C \cup C'$ is connected and $x, z \in C \cup C'$, so $x \sim z$

   We can show this last statement with the previous proposition (for fun!); define $f : C \cup C' \to \mathbb{R}$ continuous; we know that $f|_C$ has image that is an interval I and $f|_{C'}$ has image that is an interval I', so the image of f has interval $I \cup I'$, which is an interval since $f(y) \in I, I'$.

Definition 5.9. The equivalence classes of $\sim$ are called **connected components**.

Example 5.10. Let $X = \{\frac{1}{n} \mid n \in \mathbb{N}\} \cup \{0\} \subseteq \mathbb{R}$.

The connect components are all the singletons, but X does **not** have the discrete topology since $\{0\}$ is not open. Hence connected components are not necessarily open.

Example 5.11. Similarly, the Cantor set.

6 09.11.23 - Neighborhoods

Proposition 6.1. Let X be a topological space.

(i) X is a disjoint union of its connected components

(ii) Every connected $C \subseteq X$ is contained in a single component (because equivalence relations partition)

(iii) The components are closed and connected

Proof. (iii):

Connected: Let C be a component and $x \in C$. Then $\bigcup_{\substack{x \in C' \\ C' \text{connected}}} C'$ is C by transitivity and conected since the union of intersecting connected sets is connected.

Closed: Let $A \subseteq X$ be connected. Then $\overline{A} = \bigcap_{\substack{C \supseteq A \\ C \text{closed}}} C$ is also connected.

Let U, V be open in $\overline{A}$ such that $U \cap V = \emptyset$ with $\overline{A} = (U \cap \overline{A}) \cup (V \cap \overline{A})$. Then $A = (U \cap A) \cup (V \cap A)$, both of which are open in A.

Because A is connected, $U \cap A$ or $V \cap A$ is empty; WLOG, let $U \cap A = \emptyset$, then $A = V \cap A$, and $U \cap \overline{A}$ is disjoint from A.

Thus U^C is a closed set that contains A, thus $\overline{A} \subseteq U^C$, which implies that $\overline{A} \cap U = \emptyset$, which implies that $U = \emptyset$. □

In short, if any set is connected, its closure is also connected.

Proposition 6.2. Let X, Y be connected spaces. Then $X \times Y$ is connected.

Proof. Since Y is connected, (x, y) and (x, y') are in the same connected set C, and similarly, (x, y') and (x', y') are in a connected component C'. Then $(x, y) \in C \cup C'$ and $(x', y') \in C \cup C'$, so $X \times Y$ is a singular connected component and thus connected. □

Note: we need that the union of connected components is connected.

Question. If $X_j, j \in J$ is connected, is $\prod_{j \in J} X_j$ connected?

A: Yes.

Definition 6.3. Let X be a space with $x \in X$. Then $N \subseteq X$ is a **neighborhood** of x if $x \in N$ and there is $U \subseteq X$ open with $x \in U$ and $U \subseteq N$.

Remark. Neighborhoods are not necessarily open.

Definition 6.4. A **neighborhood basis** or a **neighborhood filter** of x is a set B of neighborhoods of x such that for all neighborhoods N of x, there exists $A \in B$ such that $A \subseteq N$.

Example 6.5. Let (X, d) be a metric space Then $B = \{B_\varepsilon(x) \mid \varepsilon > 0\}$ is a natural neighborhood basis of x.

This may be used to obstruct *metrizilability*, that is, to decide whether the topology is induced by the metric.

Example 6.6. $[0,1]^{\mathbb{R}}$ is not metrizable because its open sets are too large to give a countable basis, ordered by inclusion.

Given a countable neighborhood basis, we can simply take the progressive intersections to order it, i.e. $\{p_1, p_1 \cap p_2, p_1 \cap p_2 \cap p_3, \cdots\}$.

Definition 6.7. X is **locally connected** at $x \in X$ if x has a neighborhood basis of connected sets.

Proposition 6.8. If X is locally connected, then the components of X are closed and open.

Proof. Recall that connected components are closed. It hence suffices to show that the components of X are open.

Let C be a component of X with $x \in C$. Let B be a neighborhood basis at x of connected sets, then $\bigcup B \subseteq C$. There is an open $U, x \in U$ with $U \subseteq \bigcup B \subseteq C$, so C is a neighborhood of every $x \in C$, hence $C = \bigcup_{x \in C} U_x$ is open.

$\square$

7 09.13.23 - Path Connectedness

Definition 7.1. For spaces $X_j, j \in J$ and $X = \prod_{j \in J} X_j$, the **box topology** is generated by $B = \{\prod_{j \in J} U_j \mid U_j \subseteq X_j \text{open}\}$.

If j is finite, then the box topology **is** the product topology.

Why is this natural-looking topology weird, and why don't we use it?

Proposition 7.2. $\mathbb{R}^{\mathbb{N}}$ with the box topology is not connected.

Proof. Let $S = \{x \in \mathbb{R}^{\mathbb{N}} \mid x \text{is a bounded sequence}\}$ and $T = \{x \in \mathbb{R}^{\mathbb{N}} \mid x \text{is an unbounded sequence}\}$.

Clearly, $S \cup T = \mathbb{R}^{\mathbb{N}}$ and $S \cap T = \emptyset$ with $S, T \neq \emptyset$.

Now, S is open: for $(x_n)_n \in \mathbb{R}^{\mathbb{N}}$, the set $\prod_{j \in \mathbb{N}}(x_j - 1, x_j + 1)$ is open in the box topology and in S, $(x_n)_n$ is in this set. T is open for the same reason. □

Proposition 7.3. The infinite product of connected spaces is connected with regard to the product topology.

Proof. Suppose U, V are disjoint open sets in $\prod_{j \in J} X_j$ (with the product topology) with $U \cup V = \prod_{j \in J} X_j$. Only finitely many components actually matter. In this way, we can reduce it to the finite case. □

This shows that the box topology is not natural.

Proposition 7.4. In the product topology, $f : \mathbb{R} \to \mathbb{R}^{\mathbb{N}}, x \mapsto (x, x, \cdots)$ is continuous.

Proof. By the definition of initial topology, we have to verify that $\prod_j \circ f$ is continuous. They are, because they are the identity, so f is continuous. □

Proposition 7.5. This map is *not* continuous in the box topology.

Proof. If f were continuous, take $(0, 0, 0, \cdots)$; put an open sequence around it, then its preimage must be continuous. But then taking shrinking open intervals, $f^{-1}(\prod_{n=1}^{\infty}(-\frac{1}{n}, \frac{1}{n})) = \{0\}$ is not open. □

Definition 7.6. A **path** is a continuous map $\gamma : [0, 1] \to X$.

Definition 7.7. A space X is **path-connected** if for all $x, y \in X$, there is a path $\gamma(0) = x$ and $\gamma(1) = y$.

Definition 7.8. A space is **locally path-connecfted** if every point $x \in X$ has a neighborhood basis of path-connected sets.

Definition 7.9. Define the equivalence relation $x \sim y$ if there is a path from x to y. The equivalence classes are then called **path components**.

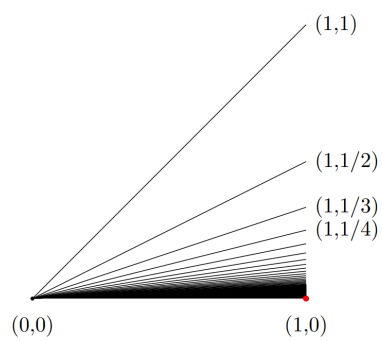
Theorem 7.10. If X is path-connected, then X is connected.

Proof. Let $x, y \in X$ be arbitrary. Since X is path-connected, there is a path $\gamma : [0,1] \to X$, $\gamma(0) = x$, $\gamma(1) = y$. Since γ is continuous and $[0,1]$ is connected, $\gamma([0,1])$ is also connected. Then x and y are in the same component, thus there is only one component and X is connected. $\square$

There are spaces that are connected but not path-connected, so path-connectedness is a stronger condition.

Example 7.11. The **deleted infinite broom**,

$$X = \bigcup_{n=1}^{\infty} [(0,0), (1, \frac{1}{n})] \cup \{(1,0)\}.$$



Proof. $X \setminus \{(1,0)\}$ is path connected by going to $(0,0)$, thus $X \setminus \{(1,0)\}$ is connected.

Fact. $A \subseteq Y$ is connected and $A \subseteq B \subseteq \overline{A}$, then B is connected.

Since $(1,0) \in \overline{X \setminus \{(1,0)\}}$, the space is connected.

However, there is no path that connects $(1,0)$ to any other point in X. Let $\gamma : [0,1] \to X$ be a path with $\gamma(0) = (1,0)$. We need to show that γ is constantly equal to $(1,0)$.

Let $T = \sup\{t \mid \gamma(s) = (1,0) \forall s \in [0,t]\}$. We want to show that $T = 1$. Since $[0,1]$ is connected, we thus need to show that if $\gamma(t) = (1,0)$ then γ remains at $(1,0)$ on a small interval around t.

γ continuous implies that there is a $\delta > 0$ such that $\forall s \in (0,\delta)$, $|\gamma(t+s) - (1,0)| < \frac{1}{2}$. Define $S : \{(x,y) \in \mathbb{R}^2, x > 0\} \to \mathbb{R}, (x,y) \mapsto \frac{y}{x}$. S is continuous, so $S \circ \gamma$ is continuous. Since $S \circ \gamma : [0,1] \to \{\frac{1}{n} \mid n \in \mathbb{N}\} \cup \{0\}$, S is a path, and because the components of Y are singletons, the path $S \circ \gamma$ is constant.

We know $s \circ \gamma(0) = 0$, thus $s \circ \gamma(t) = 0$ for all $t' < t + \delta$. Thus since $[0, 1]$ is connected, this implies that $\gamma(t') = (1, 0)$ for all $t \in [0, 1]$.

$\square$

8 09.15.23 - Open Covers

Proposition 8.1. Let X be locally path-connected. Then the path components of X are clopen and the same as the connected components.

Proof. Let C be a path component of X and $x \in C$. There is a path-connected neighborhood U of x. Then $x \in U \subseteq C$, so C is open.

Since each path component is the complement of the union of all other path components, it is also closed.

Since path components are connected, each connected component is a union of path components. But these are clopen, so there is only one path component in each component. □

Definition 8.2. Let X be a space, then $\mathcal{C}$, a set of subsets of X, is called a **cover** if $X = \bigcup_{U \in \mathcal{C}} U$.

Definition 8.3. $\mathcal{C}$ is an **open cover** if $\mathcal{C}$ is a cover and every $U \in \mathcal{C}$ is open.

Definition 8.4. A cover $\mathcal{C}' \subseteq \mathcal{C}$ is a **subcover**.

Definition 8.5. X is **compact** if every open cover of X has a finite subcover.

Example 8.6. X with the discrete topology is compact if and only if $|X| < \infty$.

Take the singletons; a subcover is finite only if there is no injection into a proper subset.

Example 8.7. Any closed and bounded set in $\mathbb{R}^n$ (Heine-Borel).

Example 8.8. $\{0,1\}^{\mathbb{N}}$ is compact because it embeds into $\mathbb{R}$ as a closed and bounded set (also look at: Cantor set).

Example 8.9. $\{0,1\}^{\mathbb{R}}$ is also compact if you "believe the right axioms of set theory".

Definition 8.10. A space X is Hausdorff if for any $x \neq y \in X$, there are open sets $U, V \subseteq X$ such that $x \in U$, $y \in V$, and $U \cap V = \emptyset$.

Example 8.11. Every metric space is clearly Hausdorff.

All topologies on a given set is a partially-ordered set (poset) that varies between the coarsest, $\{\varnothing, X\}$ and the finest, 2^X. The latter part is Hausdorff, and the former is compact. Hence essentially, Hausdorffness and compactness lie on different "ends" of this spectrum, with the set of topologies that are Hausdorff and the set of topologies that are compact possibly but not necessarily intersecting.

Proposition 8.12.

(i) A closed subspace of a compact space X is compact.

(ii) A compact subset K of a Hausdorff space X is closed.

(iii) If X is compact and $f : X \to Y$ is continuous, then $f(x)$ is compact.

Proof.

(i) Let $\mathcal{C}$ be an open cover of A. Then let $\tilde{\mathcal{C}} = \{U \subseteq X \mid U \cap A \in \mathcal{C}, U \text{ open}\}$. $\tilde{\mathcal{C}} \cup \{X \setminus A\}$ is an open cover of X (note that $X \setminus A$ is open), so it has a finite subcover $\tilde{\mathcal{C}}' \subseteq \tilde{\mathcal{C}}$. Hence $\{U \cap A \mid U \in \tilde{\mathcal{C}}'\}$ is a finite subcover of $\mathcal{C}$.

(ii) Let X be Hausdorff, $K \in X$ compact. Let $x \in X \setminus C$. For every $y \in C$, find U_y neighborhood of x and V_y neighborhood of y with $U_y \cap V_y = \emptyset$ (by Hausdorff). Furthermore, we can ensure $\overline{U_y} \cap V_y = \emptyset$.

$\{(\overline{U_y})^C \mid y \in K\}$ is an open cover of K, so it has a finite subcover $(\overline{U_{y_1}})^C, \cdots, (\overline{U_{y_n}})^C$. But then $\bigcap_{i=1}^n (\overline{U_{y_i}})^C$ is open, contains x, and is disjoint from K.

(iii) Take an open cover $\mathcal{C}$ of $f[X]$. Then $\mathcal{C}' = \{f^{-1}[U] \mid U \in \mathcal{C}\}$ is an open cover of X. As X is compact, there is a finite subcover $\mathcal{C}''$ of $\mathcal{C}'$. Now, consider the corresponding sets in $\mathcal{C}$, $\{U \mid f^{-1}[U] \in \mathcal{C}''\}$. This is a finite subcover.

$\square$

9 09.18.23 - Sequences

Theorem 9.1. Let X be compact, Y be Hausdorff, and $f : X \to Y$ be continuous and bijective. Then f is a homeomorphism.

Proof. We have to show that $U \subseteq X$ open implies that $f[U] \subseteq Y$ open.

Since f is a bijection, we show that $C \subseteq X$ closed implies that $f[C]$ closed.

Since C is closed, X is compact, and $C \subseteq X$, C is compact; this implies that $f[C]$ is compact, and because Y is Hausdorff, $f[C]$ is closed. □

Proposition 9.2. If τ_1 and τ_2 are topologies on X, $\tau_1 \subseteq \tau_2$ with τ_2 compact and τ_1 Hausdorff, then $\tau_1 = \tau_2$.

Proof. The map $f : (X, \tau_2) \to (X, \tau_1), x \mapsto x$ is continuous. By the theorem, f^{-1} is continuous, so $\tau_1 = \tau_2$. □

Question. Is there any non-Hausdorff X where compact sets are closed?

You need to construct an infinite set; any finite set would be Hausdorff. First we consider the other separability axioms.

Theorem 9.3. Let (X, d) be a compact metric space. There is no isometry to a proper subset.

Proof. Suppose $f : X \to X$ is an isometry (this means that $\forall x, y \in X$, $d(x, y) = d(f(x), f(y))$) with $f[X] \subsetneq X$.

Pick $x_0 \in X \setminus f[X]$. Since $f[X]$ is compact and closed, there is an $\varepsilon > 0$ such that $d(x_0, y) \leq \varepsilon$ for all $y \in f[X]$. Define $x_1 = f(x_0)$. As f is an isometry, $d(y, x_1) \geq \varepsilon$ for all $y \in f[f[X]]$. So $B_\varepsilon(x_1) \cap f[f[X]] = \emptyset$. Inductively, we can choose $x_n = f(x_{n-1})$ and $\varepsilon > 0$ with $B_\varepsilon(x_n) \cap f[f[\cdots f[X]]] = \emptyset$. Then this provides us with a sequence $x_0, x_1, x_2, \cdots \in X$ such that $d(x_i, x_j) \geq \varepsilon$ if $i \neq j$. Thus X is not compact. □

Definition 9.4. A metric space (X, d) is **totally bounded** if $\forall \varepsilon > 0$, there is a finite cover of X by ε-balls $B_\varepsilon(X)$.

Definition 9.5. $(x_n)_n : \mathbb{N} \to x$ is called a **Cauchy sequence** if

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n, m \geq N \quad d(x_n, x_m) < \varepsilon$$

Note. Cauchy is a metric notion only because of symmetry. It's a topological notion if you replace it with neighborhoods of x; if a space is non-Hausdorff, you can converge to multiple points.

Definition 9.6. $(x_n)_n : \mathbb{N} \to x$ **converges** to x if

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n \geq N \quad d(x, x_n) < \varepsilon$$

Definition 9.7. X is **complete** if every Cauchy sequence converges.

Theorem 9.8. Let (X, d) be a metric space. Then X is compact if and only if X is complete and totally bounded.

Example 9.9. Consider $X = \{0, 1\}^{\mathbb{N}}$ with $d : X \times X \to [0, \infty)$ via

$$((x_n)_n, (y_n)_n) \mapsto \sum_{k=1}^{\infty} \frac{|x_n - y_n|}{2}$$

inducing the product topology. It is also complete and totally bounded (exercise!), thus $\{0, 1\}^{\mathbb{N}}$ is compact.

10 09.20.23: Sequences cont.

Example 10.1. $\mathbb{R}$ with the co-countable topology, i.e.

$$\tau = \{\mathbb{R} \setminus A \mid A \text{ a countable or } A = \mathbb{R}\}$$

is non-Hausdorff and compact sets are closed.

The closed sets are exactly the countable sets on all of $\mathbb{R}$, and the compact sets are exactly the finite sets, while τ is not Hausdorff.

The induced topology on any countable subset is discrete: take any countable C and $x \in C$, then notice $\{x\} = C \cap (\mathbb{R} \setminus (C \setminus x))$ (notice that $C \setminus x$ is countable), so C has the discrete inducd topology.

Theorem 10.2. Let (X, d) be a metric space. Then X is compact if and only if X is complete and totally bounded.

Proof. $\Rightarrow$: For $\varepsilon > 0$, let $C = \{B_\varepsilon(x) \mid x \in X\}$. C has a finite subcover, thus X is totally bounded.

Now let $(x_n)_n$ be a Cauchy sequence in X that does not converge, i.e. $\forall x \in X, \exists \varepsilon > 0$ such that infinitely many $x_n \notin B_{\varepsilon_x}(x)$.

$(x_n)_n$ being Cauchy implies that there exists $N \in \mathbb{N}$ such that $\forall n, m \geq N$, $d(x_n, x_m) < \frac{\varepsilon_x}{2}$.

Let x_k be a sequence element with $k \geq N$, $x_k \notin B_{\varepsilon_x}(x)$. Then $\forall n \geq k$, $d(x_n, x_k) < \frac{\varepsilon_x}{2}$, so $x_n \notin B_{\varepsilon_x/2}(x)$ since $d(x_k, x) > \varepsilon_x$, so $\mathcal{C} = \{B_{\varepsilon_x/2}(x) \mid x \in X\}$ has no finite subcover. Hence X is complete.

$\Leftarrow$: Suppose X is not compact, then let $\mathcal{C}$ be an open cover of X without a finite subcover. Cover X by $\{B_1(x) \mid x \in X\}$.

X totally bounded means that finitely many $B_1(x)$ cover all of X. THus there is $B_1(x_1)$ that is not covered by finitely many elements of $\mathcal{C}$. Cover $B_1(x_1)$ with $\{B_{\frac{1}{2}}(x) \mid x \in B_1(x_1)\}$.

There are finitely many $B_{1/2}(x)$ that cover $B_1(x)$. Then there is some $B_{1/2}(x_2)$ that is not covered by finitely many elements of $\mathcal{C}$.

Proceed inductively: once x_n is constructed, cover $B_{1/2^n}(x_n)$ by $\{B_{1/2^{n+1}}(x) \mid x \in B_1(x)\}$. $B_{1/2^{n+1}}(x_{n+1})$ is not covered by finitely many elements of $B_{1/2^{n+1}}(x_{n+1})$, thus we construct a sequence $(x_n)_n$.

$(x_n)_n$ is a Cauchy sequence. For $\varepsilon > 0$, find $N \in \mathbb{N}$ with $\frac{1}{2^N} < \varepsilon$. Notice $\sum_{n=N+1}^{\infty} \frac{1}{2^N} = \frac{1}{2^N}$, thus $d(x_n, x_m) \leq \frac{1}{2^N}$ for $n, m \geq N$.

X complete implies that there exists some $x \in X$ such that $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$, $\forall n \geq N$, $d(x, x_n) < \varepsilon$. There is $U \in \mathcal{C}$ with $x \in U$. There is $\delta > 0$ such that $B_\delta(x) \subseteq U$.

For x_n with $d(x_n, x) < \delta/2$ and $\frac{1}{2^n} < \frac{\delta}{2}$, we have that $B_{1/2^n}(x_n)$ is covered by U. $\square$

Definition 10.3. Let X be a topological space. $D \subseteq X$ is called **dense** if $\overline{D} = X$.

Definition 10.4. X is **separable** if there is a countable dense set.

Example 10.5. $\mathbb{R}$ with the standard topology is separable because $\overline{\mathbb{Q}} = \mathbb{R}$.

Example 10.6. $\mathbb{R}$ with the co-countable topology is *not* separable, because every countable set is contained in a closed set $\subseteq \mathbb{R}$.

Theorem 10.7. (X, d) is homeomorphic to a totally bounded space if and only if X is separable.

Remark. There exist separable spaces that are not metrizable, e.g. $\mathbb{R}$ with the cofinite topology.

11 09.22.23

Theorem 11.1. Let (X, d) be a metric space. Then X is separable if and only if X is homeomorphic to a totally bounded space.

Proof. $\Leftarrow$: For every $n \in \mathbb{N}$ over X with finitely many balls of radius $\frac{1}{n}$, say

$$X = \bigcup_{i=1}^{k(n)} B_{\frac{1}{n}}(x_i^{(n)}),$$

let

$$D = \{x_i^{(n)} \mid n \in \mathbb{N}, i \in \{1, 2, \cdots, k(n)\}\}$$

be countable.

Claim: $\overline{D} = X$

Fix $x \in X$. Let $\varepsilon > 0$. We have to show that $B_\varepsilon(x) \cap D \neq 0$. Let $n \in \mathbb{N}$ with $\frac{1}{n} < \varepsilon$. Now, x is in some $B_{1/n}(x_i^{(n)})$, which implies that $x_i^{(n)} \in B_\varepsilon(x) \cap D$.

$\Rightarrow$: If X is separable, then X embeds into $[0,1]^{\mathbb{N}}$ with the metric

$$d_p : [0,1]^{\mathbb{N}} \times [0,1]^{\mathbb{N}} \to [0, \infty)$$

$$((x_n)_n, (y_n)_n) \mapsto \sup_{n \in \mathbb{N}} \frac{|x_n - y_n|}{n}$$

$\square$